

Generation Interconnection System Impact Study Report

For Queue Project Q19-1241

Boeing Renton 2 100 kV

July 26, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1241. The Interconnection Customer proposes a 105.4 MW solar photovoltaic facility interconnecting at the Boeing Renton 2 100 kV substation in the Indiana area. The request was received on July 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 309 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,137,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hassen Cassel / Muldoon Mill Substation 100 KV	Wayne Trace / Darlington Road 100 KV	165.5	107	5.4
Hassen Cassel / Darlington Rd 100 KV	Wayne Trace / Darlington Rd 100 KV	139.5	118	5.4
S Whitehall / Roseau County 100 KV	John S Rainey / Trier Substation 100 KV	114.4	131	11.7
Pt A / Lake Marion 100 KV	Dekalb Substation / Richland 500 KV	112.9	172	5.7
Barret / Pt A 100 KV L	Richland / Dekalb 500 KV	112.5	191	5.7
Monroeville / Woodburn 100 KV	Richland Substation / Dekalb 500 KV	104.5	230	5.6
Sage / Woodburn 100 KV line	Dekalb / Richland 500 KV	103.6	244	5.6
Monte Del Estado 6303 / Sage 100 KV	Dekalb / Richland 500 KV Line	102.6	264	5.6
Baret / Roseau County 100 KV L	Richland / Dekab 500 KV line	101.9	200	10.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hassen Cassel / Muldoon Mill Substation 100 KV	Rebuild line with higher-capacity conductor	\$2,300,000	30% (\$689,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hassen Cassel / Darlington Rd 100 KV	Rebuild line with higher-capacity conductor	\$3,202,000	30% (\$959,000)
Roseau County Substation / S Whitehall 100 KV	Rebuild line with higher-capacity conductor	\$8,639,000	41% (\$3,568,000)
Pt A / Lake Marion Substation 100 KV	Rebuild line with higher-capacity conductor	\$4,902,000	10% (\$516,000)
Barret / Pt A 100 KV L	Rebuild line with higher-capacity conductor	\$12,466,000	10% (\$1,312,000)
Monroeville / Woodburn 100 KV line	Reconductor line	\$2,423,000	28% (\$685,000)
Sage / Woodburn 100 KV	Reconductor line	\$1,119,000	28% (\$316,000)
Sage / Monte Del Estado 6303 100 KV line	Reconductor line	\$1,017,000	28% (\$288,000)
Baret / Roseau County 100 KV L	Reconductor line	\$2,326,000	35% (\$804,000)
	Total allocated to project		\$9,137,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.